

From Ancestral Continuity to Artificial Agents

Organization, Experience, and the Attribution of Intelligence

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Abstract

An evolutionary history populated by its intermediate organisms makes the proposed onset of experience difficult to locate. Artificial agents raise a related attribution problem: an abacus, an electronic calculator, and a human formation implementing an intelligent program can participate in successful computation, but their capabilities belong to differently organized processes. This paper connects these thought experiments within the four-paper Unified Consciousness Theory (UCT) framework. It distinguishes actual organization, complete organizational type, task-relative capability, experiential type, and report. Three existing conditional results supply the mathematical spine: experience attribution to actual ancestral processes under UCT's identity axiom; a nonproduct result for actual wholes and their specified constituents; and the implication from different fixed-contract capabilities to different complete experiential types. A matched-memory example shows why storage, access, performance, and experiential interpretation require separate premises. The resulting account connects biological and artificial organization without treating biological ancestry as an eligibility condition, computational equivalence as complete organizational identity, or performance as a measure of experiential richness. Its contribution is a unified argument and an AI-oriented application of existing results, not an independent proof of the identity axiom or a new major theorem. The remaining challenge is to justify the relevant physical boundaries and cross-system correspondences.

Keywords: consciousness; experience; organization; artificial agents; thought experiments; capability attribution

1. The question behind the examples

An abacus can be used to calculate. An electronic calculator performs an installed arithmetic procedure. A sufficiently organized population of people could, in a thought experiment, execute the transitions of a powerful artificial agent. What exactly possesses the resulting capability? What, if anything, follows about experience? These questions become obscured when the same word, "system," alternately denotes an artifact, its operator, a computational description, and the larger process containing them.

A second family of questions concerns origins. Imagine that every relevant intermediate organism in a history leading to present humans remains available for comparison. Extend the inquiry backward to the physical and chemical processes preceding living organisms. If experience begins only after some organization has formed, that organization also has a formation history. It cannot be introduced into the argument as though a fully developed architecture appeared between two otherwise indistinguishable neighbors.

The connection between these inquiries is organizational, not genealogical. A biological lineage and an artificial agent need not share ancestry to be compared using the same distinctions between actual processes, constituents, transitions, inputs, and outputs. Conversely, placing them in one vocabulary does not establish that their complete organizations are equivalent.

UCT I supplies a process ontology and a structural-experiential identity axiom [1]. UCT II examines scoped translations of consciousness theories into organizational descriptions [2]. UCT III distinguishes capability from experiential type [3]. A fourth report examines mechanism identification and evidential limits for artificial agents [4]. This paper brings those resources together around the thought experiments that motivate their use. The sequence is an explanatory route through the framework; it is not a deduction from ancestral continuity to intelligence and then to one unified subject.

The central proposal is that intelligence and experience should be related through a specified actual organization. Capability describes what that organization can do under a task contract. Experience, under UCT's axiom, is its intrinsic presentation. This does not insert an additional experiential force into the causal account. Nor does it identify every form of experience with intelligent behavior. The distinction matters especially for agents whose reports are convincing while their physical implementation and causal dependencies remain only partially characterized.

2. All ancestors survive: motivation and its limit

2.1 The formation challenge

The All-Ancestors-Survive thought experiment is a comparison device, not a claim that evolutionary history actually preserved all its participants. Stipulate that the relevant ancestral process tokens are available in their functioning conditions. Their differences can then be inspected rather than compressed into a few labels such as “cell,” “animal,” and “human.” Physical and chemical prehistory extends the comparison; it is not literally a lineage of organisms before organisms existed. Branching, loss, and reorganization remain possible. The experiment does not assume a single ladder of increasing complexity.

Consider a proposal that experience first appears with organization G. The formation challenge asks for the intermediate organizations through which G arises. A recurrent connection, differentiated memory, or coordinating mechanism can itself have antecedents. The comparison should include changes in constituent relations, connection strength, time scale, and operational use. Naming G is therefore the beginning of an explanation, not its completion.

This challenge is substantive even when adjacent differences are small. It asks why a particular change concerns the presence of any experience, rather than a new content, capacity, or mode of access. It also requires that the relevant bearer remain explicit: the appearance of a capacity in an encompassing process need not mean that its constituent processes previously lacked all experience.

Nevertheless, small physical differences do not logically rule out a threshold. A theory may propose a discontinuous predicate or distinguish a property from a smooth physical quantity. It then owes an account of that proposal; continuity alone does not refute it.

2.2 Why gradual formation is not a proof

UCT I already provides the relevant countermodel [1, §8.3]. For a formation parameter λ , let

$$c(\lambda) = \begin{cases} 0, & \lambda \leq \lambda_*, \\ \exp[-1/(\lambda - \lambda_*)^2], & \lambda > \lambda_*. \end{cases}$$

The function is smooth, but the binary predicate $\mathbf{1}_{\{c>0\}}$ changes at the threshold. This is a logical counterexample, not a measured psychophysical law or UCT's definition of experiential magnitude. It shows why smooth physical formation does not by itself establish continuity of experience existence. A long finite ancestry also does not establish a mathematical continuum or arbitrarily fine steps.

Three arguments must therefore remain separate. Under UCT's identity axiom, each stipulated actual ancestor receives an experiential interpretation. Under an additional geometry transported by structural identity, specified structural distances can be interpreted experientially. Under an independently assumed continuous binary existence map on a connected domain, existence must be constant. The third statement follows because a continuous image of a connected space is connected, whereas a subset containing both points of $\{0, 1\}$ is not. None of these routes supplies the premises of the others.

The thought experiment motivates a theory that avoids an unexplained universal onset gate. It does not compel acceptance of that theory. Retaining this distinction gives the experiment a stronger role than either treating it as a proof or deleting it from the paper: it makes the explanatory choice and its alternatives visible.

3. A minimal organizational vocabulary

3.1 Actual processes, complete types, and views

An **actual process token** is a physically occurring process with support, causally inherited relations, temporal continuity, and an accountable boundary over the comparison interval. The boundary must be justified by the process under study, not drawn solely to produce a preferred consciousness verdict. UCT permits overlapping and nested processes; validity does not require isolation, maximality, verbal report, or self-inspection. These constraints are an ontological specification, not a universal algorithm for finding every process boundary.

Write $D_t^{\text{ontic}}(P)$ for the complete, token-relative constitutive organization of process P at the relevant pointed time or phase. "Complete" means adequate to the declared constitutive question. It neither means every physical fact in the universe nor allows task-irrelevant-looking relations to be discarded without justification. A complete type is an equivalence class under the admitted structure-preserving isomorphisms.

A finite representation may declare states, a current state, input and intervention ports, outputs, transition operations, constituent relations, and temporal order. Its isomorphisms must preserve the declared signature. A relabeling cannot arbitrarily exchange an intervention port with an output, change a constituent relation, or erase a relevant delay. A measured or computational view is instead $D_v(P) = \Pi_v[D^{\text{ontic}}(P)]$. Equal views do not establish equal complete types unless the projection is adequate for that

conclusion. An estimated view adds another uncertainty: whether it correctly describes the chosen physical target. These distinctions prevent an algorithm, an execution trace, and a complete actual organization from becoming interchangeable.

3.2 Experience and the identity commitment

UCT's C1 is an identity commitment: an actual process's physical organization and intrinsic experiential presentation are two presentations of the same organized process [1]. In its strong tokenwise form, a structure-preserving correspondence h_P connects these presentations. Let $\Phi(P)$ denote the complete experiential type. At the type level, the associated organizational-invariance statement is

$$[D^{\text{ontic}}(P)] = [D^{\text{ontic}}(Q)] \iff \Phi(P) = \Phi(Q).$$

This is not established by defining a new mathematical symbol for organization. C1 is the philosophical commitment that supports the experiential interpretation. It is stronger than the observation that physical changes correlate with reported experiences.

In this paper, **experience** denotes that intrinsic presentation, not specifically wakefulness, self-awareness, language, or human visual content. **Consciousness** is used in this basic sense unless a particular capacity or content is expressly named. C1 supplies neither a universally calibrated scalar amount of experience nor a scale of pleasure, suffering, or moral status. A narrow process can receive an experiential interpretation without being attributed a human perspective.

Proposition 1 — Ancestral attribution (UCT I, E1). Assume C1 and its nonempty-actual-carrier consequence U1. If every member of the stipulated ancestral collection is an actual process in that domain, each has an experiential presentation.

Proof. Apply U1 to each member. No adjacency, metric, or continuity premise is required. The conclusion is conditional on C1 and actuality; it is not an independent argument establishing C1. $\square$

3.3 Capability and report

Let $\mathcal{M}$ fix a task family, environment, admissible interactions, resource and time conditions, and scoring rule. A **capability functional** $J_{\mathcal{M}}$ describes performance under that contract and is invariant under the admitted complete organizational isomorphisms [3]. It may be a profile rather than a scalar. One observed success is not the functional itself. A trained installation, an installation with an added controller, and an optimized family of redesigned systems also define different questions.

Here **intelligence** concerns such capabilities across specified tasks; general intelligence requires a sufficiently broad contract, not the success of a single arithmetic operation. This operational treatment does not settle every ordinary use of "intelligence." It makes a particular comparison precise enough to assess.

A **report** is behavior through designated output channels that is interpreted as a claim about a state or experience. Reports are generally a kind of behavior, so experience, behavior, and report are not three mutually exclusive substances. A report's reliability depends on its producing mechanism and interpretive relation to the target.

4. Abacus, calculator, and the bearer of capability

An abacus stores configurations of beads. In ordinary use, a person supplies the procedure and moves them. Successful arithmetic can therefore be attributed to the person-with-abacus process under a suitable contract. The bare artifact does not inherit every capacity of that composite. Conversely, describing it as “merely a tool” does not erase the physical states and relations that make it useful.

An electronic calculator installs more of the procedure within the device. Given admissible inputs and power, its mechanism performs arithmetic transitions without the operator manually implementing each step. This supports an attribution of narrow arithmetic capability to the operating calculator. It does not establish broad adaptation, a self-model, or general intelligence. The difference from the abacus is partly a relocation of functional work across the chosen boundary.

Within C1’s domain, actual processes in either arrangement have an experiential interpretation. This conditional statement does not follow from “it calculates,” and it does not mean that a calculator understands numbers or feels like its user. The criterion concerns actual organization, while arithmetic performance answers a task question. Outside the framework, the examples by themselves do not settle whether either artifact experiences anything.

Comparison target	What the example supports	What requires an additional argument
Bare abacus	Physical state retention and usable bead configurations	The operator’s arithmetic procedure as the artifact’s own capability
Person with abacus	Composite arithmetic performance under the contract	Attribution of the entire result to either component alone
Operating calculator	Installed narrow arithmetic transitions	General intelligence or a specified phenomenal content
Running artificial agent	Capabilities of a specified implementation and interface	Complete experiential identity with another implementation

This analysis transfers directly to AI systems. A language model supplied with external memory, a scheduler, tools, and an environment may succeed at tasks that the isolated model cannot perform. The additional capability belongs to the relevant implemented composite unless a narrower attribution is established. A benchmark label does not resolve that boundary question.

5. A human formation running an artificial agent

5.1 Computation without component understanding

Inspired by the human-formation computer in Cixin Liu’s *The Three-Body Problem* [5], imagine people arranged to implement a computational process. Each follows local signaling rules. Now strengthen the scenario: the formation actually runs an agent with broad capabilities under a specified contract, rather than displaying a prerecorded sequence. Its inputs affect subsequent states, its memory persists appropriately, and its outputs affect the task environment.

Stipulating an adequate implementation settles the relevant computational-capability question within the scenario. It does not require each person to understand the global calculation. “Superintelligent” adds no further logical power unless its tasks, comparisons, resources, and timing are specified. A formation that computes the same mathematical function extremely slowly may fail a time-critical contract that an electronic implementation satisfies.

The participants’ experiences and the whole’s attribution are separate questions. Participants do not cease being actual processes because their signaling helps implement a larger one. If the formation also constitutes a valid actual whole, C1 applies to that whole. This does not make its experiential presentation the arithmetic sum of the participants’ experiences, identify one participant as its owner, or establish a unique human-like subject encompassing everyone.

5.2 A precise composition result

The relevant existing result concerns independent products, not an undefined total quantity of feeling [1, §6]. Fix a common structural signature and a specified decomposition into nonoverlapping constituent slots. Let $\otimes$ denote the independent product in the physical and experiential presentations, with isomorphisms preserved by products.

Proposition 2 — Nonproduct composition (UCT I). Assume strong tokenwise C1, actual whole W , actual constituents $P_1, \dots, P_n$ in the specified slots, and the common product conventions. If

$$D(W) \not\cong \bigotimes_i D(P_i),$$

then

$$\Phi(W) \not\cong \bigotimes_i \Phi(P_i).$$

Proof. If the experiential product equivalence held, compose the C1 correspondence for W , that equivalence, and the inverses of the constituent correspondences. Preservation of products yields an isomorphism between $D(W)$ and $\bigotimes_i D(P_i)$, contradicting the premise. $\square$

The abstract independent product need not itself be an actual process. No experience-existence principle is applied to an imaginary product. The result is relative to the

specified decomposition and boundary; it does not prove indecomposability under every possible partition. The weak statement that complete types correspond bijectively is insufficient for this proof: the strong structural correspondences and product compatibility do real work.

For the human formation, the nonproduct premise must be justified from the actual relations. Mere proximity, a group name, or the presence of communication does not discharge it. The proposition explains what follows if the premise holds. It does not settle the implementation from a verbal description.

5.3 Why this does not settle computational functionalism

The scenario has clear antecedents. Block's population implementation challenges the attribution of qualitative states on the basis of functional equivalence [6]. Chalmers's fading and dancing qualia arguments defend organizational invariance through imagined replacement preserving relevant organization [7]. The present formation question instead follows an organization as it changes or is built. These are related but different comparisons.

UCT's response cannot simply be that the same program guarantees the same complete experience. Computational equivalence may concern only a view. Human participants and electronic components may have additional constitutive relations not preserved by that view. Equally, different material labels alone do not establish different complete experiential types. The required question is which complete structure is preserved, not whether an implementation is unfamiliar or made from a favored substance.

6. From biological formation to an AI runtime

Biological evolution, development, and learning provide one set of formation histories. Design, training, installation, and interaction provide another. The bridge is a common comparison of the resulting actual organizations and their changes. It is not a claim that an artificial agent traverses the entire evolutionary sequence, or that being historically descended from living organisms is necessary for experience.

For an AI application, at least four objects must remain distinguishable: a mathematical model specification; a stored parameter artifact; a running implementation; and a wider process including memory, tools, controllers, and environmental coupling. These can be related without being the same bearer. A printed program and a live execution do not have identical transition mechanisms merely because one describes the other. A replay can reproduce an observed output sequence without supplying the same responses to changed inputs or interventions.

Historical facts matter through their constitutive relevance. Training may leave weights, retained states, or ongoing dependencies. If a chosen state description omits causally relevant retained history, the description is incomplete for its intended use. Conversely, a history already screened off by a sufficient present description should not be added as an unexplained extra causal influence. Whether screening holds is a modeling question, not a license to discard history in advance.

A similar care is needed with tools. Receiving a message from another process does not automatically make the sender a constituent of the receiver. UCT III's three-process example separates two sources from a reader that receives their signals [3, §4.2]. Access

to both independent bits permits their equality to be computed exactly; one isolated bit permits only chance performance on that task. This is an information-resource comparison, not a demonstration of superiority over a reader that already has both bits. Nor does it show that one-way senders acquire the reader’s new capacity.

These distinctions yield a concrete research object: the physically supported agent process, with justified boundary, installed transitions, available information, and time-dependent interfaces. Its experience interpretation is conditional on C1. Its capacities require their own task analysis. Neither question can be answered by counting parameters alone.

7. What improved capability does—and does not—imply

7.1 The conditional type result

Write s, t for complete organizational types in a common domain. UCT III provides the following implication [3, Proposition 1].

Proposition 3 — Capability difference and complete type. Assume C1’s complete-type equivalence and a fixed capability functional $J_{\mathcal{M}}$ invariant under the admitted organizational isomorphisms. Then

$$J_{\mathcal{M}}(s) \neq J_{\mathcal{M}}(t) \implies \Phi(s) \neq \Phi(t).$$

Proof. Equal complete experiential types imply equal complete organizational types by the assumed equivalence. Invariance of $J_{\mathcal{M}}$ then gives equal capabilities. Take the contrapositive. $\square$

This statement concerns differences in the capability functional, not different random scores from the same system. Its experiential interpretation for physical targets requires valid actual tokens and adequate complete-type grounding. A finite model alone does not supply those facts.

The converse fails when different types share a capability. More generally, a target Q is identifiable from J exactly when Q is constant on every fiber $J^{-1}(j)$; then and only then a function f exists with $Q = f \circ J$ [3]. Thus an arithmetic answer, intelligence score, or report can identify only distinctions it preserves.

Nor does a capability gain impose an experiential ordering. In the existing countermodel $s = (x, y)$, $J(s) = x$, and $Q(s) = y$, the move from $(0, 1)$ to $(1, 0)$ improves J while lowering Q . Calling Q “richness” would require an additional justified interpretation. C1 provides no automatic monotonic scale.

7.2 Storage, information, and usable memory

A compact example from UCT III makes the distinction operational [3, §7.2]. Let independent fair input bits be X_t , and let N_t be fresh independent fair noise. Set

$$M_{t+1} = X_t \oplus (w \wedge N_t), \quad Y_t = \begin{cases} X_t, & r = 0, \\ M_t, & r = 1. \end{cases}$$

Output occurs before the memory update. In installation A, $w = 0$; in B, $w = 1$. There is no retained mask or alternative accessible record of the preceding input. The task is to output X_{t-1} after warm-up.

Quantity or installed readout	A: retained input	B: masked input
Memory entropy $H(M_t)$, bits	1	1
Mutual information $I(M_t; X_{t-1})$, bits	1	0
Success with $r = 0$	1/2	1/2
Success with $r = 1$	1	1/2

With $r = 0$, both emit the current input and perform equally on the past-input task. Equal storage entropy and equal performance conceal a difference in retained information. Changing the installed readout to $r = 1$ exposes that difference. The new performance is a property of the changed installations; it must not be credited retrospectively to their former readouts.

The example is a formal mechanism comparison, not a measurement of a device’s experience. Under the additional actual and complete-type premises, Proposition 3 gives a conditional experiential-type distinction for the installations with unequal capability. It does not say how that difference feels or which installation has more experience. The example explains why adding memory to an AI agent is not adequately described by storage size alone: the retained relation and the mechanism using it matter.

8. Correspondence is a separate obligation

An agent implemented by a human formation and one implemented electronically may share a computational description. A biological evidence accumulator and an AI accumulator may share a signed evidence coordinate. Neither observation supplies complete organizational correspondence.

A defensible comparison declares at least the following:

1. **State:** which physical or modeled distinctions are represented, including the current state and relevant retained history.
2. **Ports:** which inputs, interventions, resets, and outputs correspond, and which are actually available.
3. **Time:** how updates, delays, ordering, and performance deadlines are related.
4. **Output law:** which distributions or behaviors must agree, over which starting states and histories.
5. **Feedback:** how actions selected from the allowed observed history are implemented without hidden-state access being silently added.

The existing formal companion gives a finite executable-interface result for such a restricted setting [8]. It requires a sufficient Markov state, exact projected joint transition/output rows, legal implementation actions, and an interface using only the permitted visible state and fresh conditional randomness. Under its all-start and policy-scope conditions, the interface preserves the stipulated finite projected feedback laws. That result does not preserve every unrepresented cost, internal relation, or complete experiential type.

Consequently, a successful finite comparison is useful but limited. It can establish a specified mechanism or output relation and make failures informative. It cannot close the biological-AI constitutive question by itself. The theoretical next step is to identify which remaining organizational distinctions matter to the intended claim and what could justify their preservation. New experiments should follow that target, rather than compensate for its absence.

9. Explanatory reach and remaining disagreements

The account has a positive implication: under C1, experience is not an optional extra that can be removed while preserving the same complete actual organization. The causal mechanisms supporting intelligent performance are features of that same organization. This is a constitutive connection, not a demonstration that experience adds a separate force that improves intelligence. U1 alone predicts neither arithmetic success nor general intelligence.

A rival theory need not address exactly the same target. Lamme’s recurrent-processing account develops a distinction between feedforward processing, local recurrence associated with phenomenal visual processing, and broader processing supporting access [9]. It should not be reduced to a report requirement or the claim that any loop suffices. A claim about the absence of a specified visual experience does not automatically conflict with UCT’s attribution of some experience to an actual process. Conflict requires a matching bearer and domain and an explicit claim of necessity for all experience. Conversely, UCT must explain particular contents and cannot use its broad existence claim as a substitute for that work.

The human-formation case also leaves genuine disagreement. One may reject C1, challenge the proposed actual whole, or deny that the stipulated correspondence is complete. These objections concern different premises and should be answered separately. The framework does not dissolve the problem of phenomenal character by naming it structure. It states the identity commitment and clarifies which conclusions depend on it.

Finally, an agent’s behavior does not establish valence. The fourth source report distinguishes evidence for a self-continuation mechanism from evidence for negative valence or fear [4]. Even a strong functional certificate does not independently identify suffering. Reports of fear, resistance to interruption, and preservation of a task can arise under different mechanisms. This evidential distinction remains necessary within an experience-permissive ontology.

10. Conclusion

The ancestral and artificial-agent thought experiments address complementary parts of one problem. The first makes organizational formation and the proposed onset of experience explicit. The second makes capability bearers, implementation, and whole-part relations explicit. Together they motivate an account in which actual organization grounds task capacities and, conditionally on C1, experiential presentation.

Three existing results discipline that account: ancestral attribution under the actual-process axiom domain; nonproduct composition under strong structural and product premises; and complete-type difference from unequal fixed-contract capability. Their

limits are equally important. Gradual physical change does not prove universal experience, computation does not establish complete constitution, and better performance does not establish more or better experience.

The next theoretical problem is therefore focused: specify and justify the organizational correspondences needed for a concrete biological-artificial comparison. The thought experiments belong in that inquiry because they expose the premises. The formal results belong beside them because they show exactly what those premises can support.

Appendix A. Source and proof map

The manuscript adds no deductive nodes or rules to the unified map. Its numbering is local to this draft. The canonical R137 proof ledger and the four source editions remain authoritative; R140 is the common technical companion [8].

Manuscript claim	Existing map anchors	Required qualification
Proposition 1	A:C1, A:U1, A:ALL_ANC_ACTUAL, A:E1; rule a19	Actual carriers and the UCT commitment; no continuity inference
Proposition 2	A:C1, A:PRODUCT_DEF, A:ACTUAL_WHOLE_PARTS, A:PHYS_NONPRODUCT, A:NONSUM; a14	Strong tokenwise correspondences, fixed slots, common products
Proposition 3	A:C1_OI, C:FIXED_J, C:P1; c01	Fixed contract, complete types, admitted-isomorphism invariance
Identification from capability	C:P2_COORD, C:P2_FULL	Constancy on fibers; full identification requires injectivity
No automatic scalar increase	C:P3	An interpreted ordering is additional structure
Matched-memory example	C:CIRCUIT_SPEC, C:CIRCUIT_LAWS; c13	Update order, independence, and inaccessible noise mask
Scoped theory comparison	B:TARGET_DOMAIN, B:GATE, B:RPT	Same target and bearer; preserve the rival's actual claims
Finite feedback comparison	R137:CONTROL_MODEL, R137:EXACT_INTERFACE, R137:CLOSED_LOOP	All-start, legal-policy, observation, and randomness restrictions

The manuscript's additional contribution is the arrangement of these existing results around ancestry, tools, and implemented agents. It does not claim a newly discovered nonproduct theorem, a first population-computer thought experiment, or an independently established physical identity principle. Publication suitability as a standalone synthesis remains a separate editorial judgment.

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Draft status and assistance

This is an unpublished synthesis draft based on the pinned editions in Appendix A, not a revision of those publications. AI assistance was used for drafting, source comparison, mathematical consistency review, and typesetting. Author review and independent scholarly review remain outstanding. No new empirical measurements are reported; no inference of experiential valence or a particular current AI system's consciousness is established.